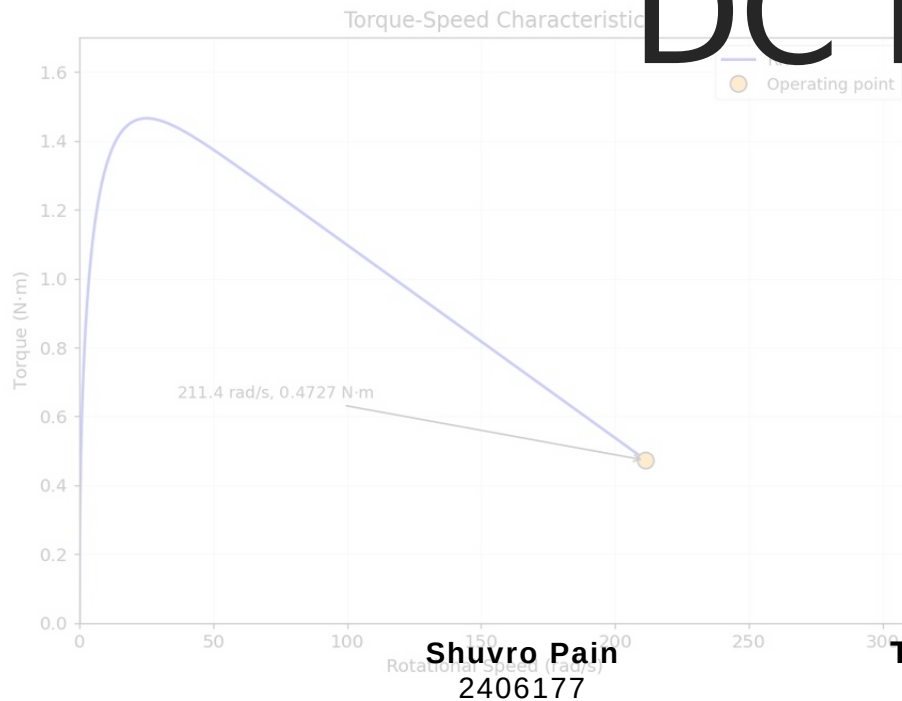


EEE 212 – Numerical Techniques Laboratory
Level-2 Term-I Section C2
Final Project Demonstration

DC Motor Dynamic Simulator



Taseen Intisar
2406188

Abdus Samiul Hasan
Sun
2406193

Nadman Wasit
2406195

1



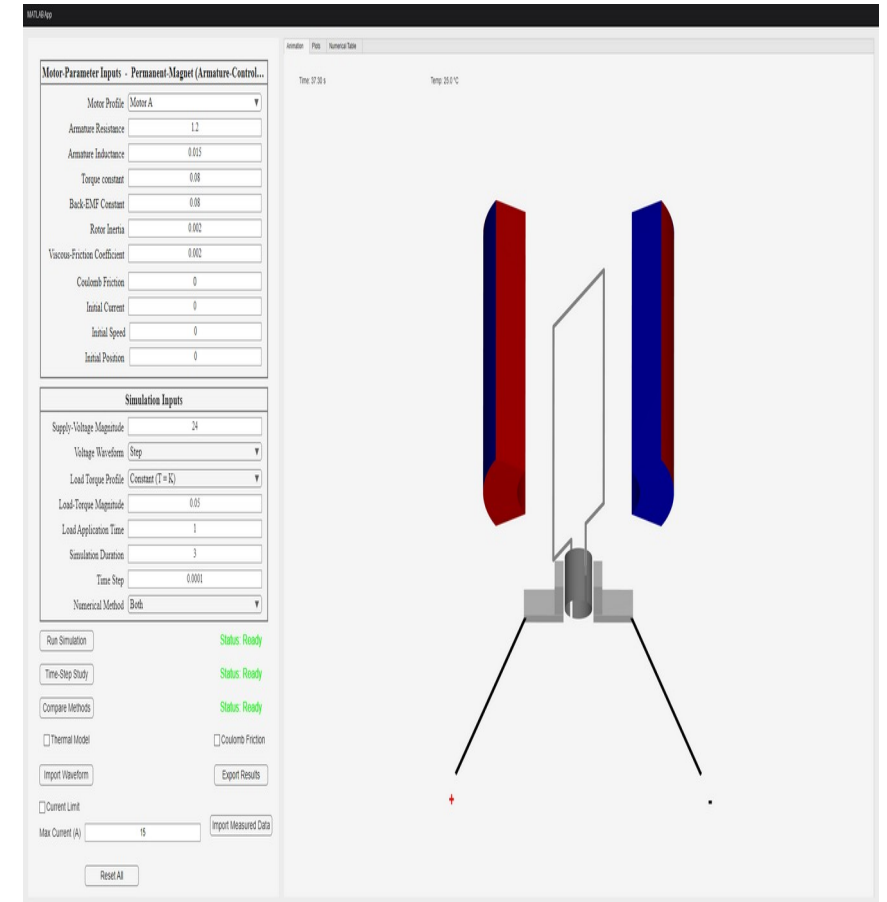
BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Outline

1. Summary
2. Introduction
3. Design
4. Implementation
5. Analysis and Evaluation
6. Future Work
7. References

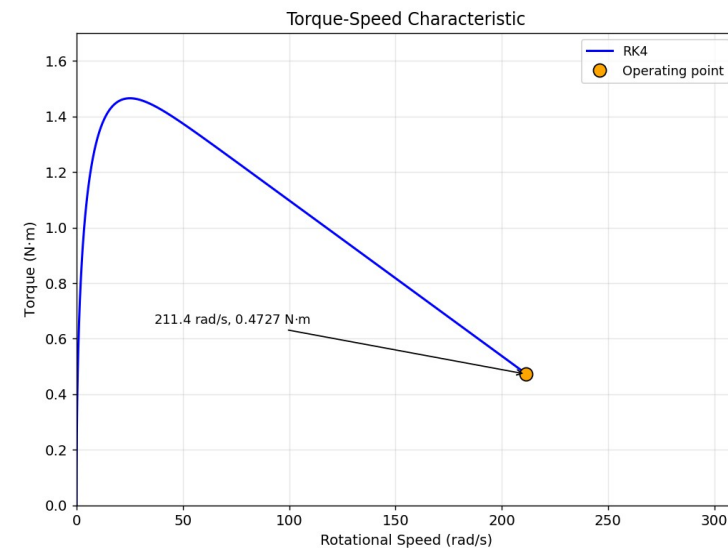
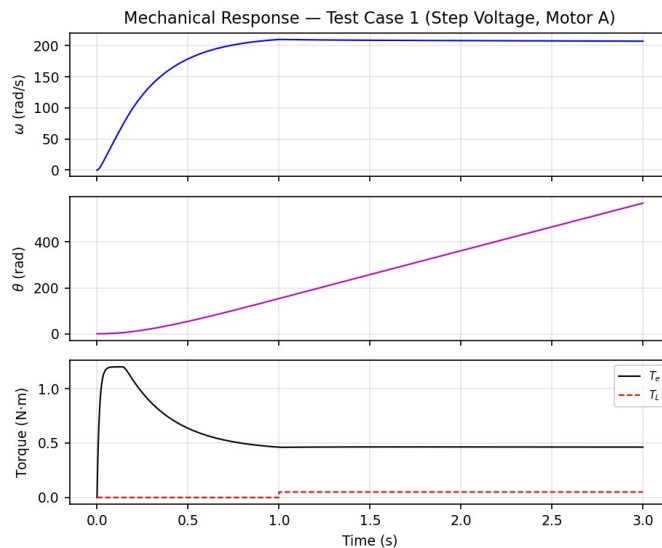
1. Summary / Abstract

- MATLAB App Designer application that simulates an armature-controlled DC motor
 - Models the coupled electrical and mechanical dynamics of the machine
- Three core states solved numerically: armature current, speed, shaft position
- Forward Euler and RK4 implemented from first principles
 - ode45 used only as a validation reference, never as the solver
- Interactive GUI with animation, plot views and a numerical results table
- Optional extensions: thermal model, current limiting, Coulomb friction, speed-dependent loads



2. Introduction

- A DC motor converts electrical energy into mechanical energy
- Current, speed and position all vary with time - it is a dynamic system
- Electrical and mechanical subsystems are coupled through back-EMF and torque
 - No closed-form solution once current limiting or a speed-dependent load is added
- Numerical integration is required to predict the transient response



3.1 Design: Problem Formulation (PO(b))

3.1.1 Identification of Scope

- **Problem**

- Formulate the coupled differential equations of an armature-controlled DC motor
- Solve them numerically and analyse the dynamic response
- Compare at least two numerical integration methods

- **Scope**

- States: armature current, angular speed, shaft position
- Variable applied-voltage waveforms and load-torque characteristics
- Forward Euler and RK4 implemented; ode45 for validation only
- Optional: thermal model, current limit, Coulomb friction, adaptive RK45, animation
- Out of scope: magnetic saturation, regenerative braking into the supply

3.1 Design: Problem Formulation (PO(b))

3.1.2 Literature Review

- Chapman, Electric Machinery Fundamentals - DC motor operating principles
- Chapra & Canale, Numerical Methods for Engineers - Euler and Runge-Kutta methods
- MathWorks ODE documentation – ode45 for adaptive time-step integration
- MathWorks App Designer documentation - GUI components, callbacks, axes and tables

3.1 Design: Problem Formulation (PO(b))

3.1.3 Formulation of Problem

- Kirchhoff voltage law on the armature circuit, Newton second law on the rotor
- Coupled through the back-EMF constant K_e and the torque constant K_t

$$V_a(t) = L_a \frac{di_a}{dt} + R_a i_a + e_b \quad (\text{armature circuit})$$

$$e_b = K_e \omega \quad (\text{back-EMF})$$

$$T_e = K_t i_a \quad (\text{electromagnetic torque})$$

$$J \frac{d\omega}{dt} + B \omega = T_e - T_L \quad (\text{rotor dynamics})$$

$$\frac{d\theta}{dt} = \omega \quad (\text{shaft position})$$

3.1 Design: Problem Formulation (PO(b))

3.1.4 Analysis

- Rearranged into explicit state-derivative form for the numerical solvers

State vector: $\mathbf{x} = [i_a, \omega, \theta]^T$

$$\frac{di_a}{dt} = \frac{V_a(t) - R_a i_a - K_e \omega}{L_a}$$

$$\frac{d\omega}{dt} = \frac{K_t i_a - B\omega - T_L(t, \omega)}{J}$$

$$\frac{d\theta}{dt} = \omega \quad \Rightarrow \quad \dot{\mathbf{x}} = f(t, \mathbf{x})$$

Load torque follows a torque-speed characteristic

$$T_L(t, \omega) = u(t - t_L) \cdot g(\omega), \quad \omega_{ref} = V_a / K_e$$

$$\text{Constant: } g = T_m \quad | \quad \text{Linear: } g = T_m \omega / \omega_{ref}$$

$$\text{Quadratic: } g = T_m \omega |\omega| / \omega_{ref}^2 \quad | \quad \text{Const. power: } g = T_m \omega_{ref} / |\omega|$$

3.2 Design: Numerical Methods

- Two explicit fixed-step integrators written directly, not called from a toolbox
- Both advance the same shared state-derivative function, so the model is identical

Forward Euler : $\mathbf{x}_{n+1} = \mathbf{x}_n + h f(t_n, \mathbf{x}_n) \quad \mathcal{O}(h)$

RK4 : $k_1 = f(t_n, \mathbf{x}_n), \quad k_2 = f(t_n + \frac{h}{2}, \mathbf{x}_n + \frac{h}{2}k_1)$

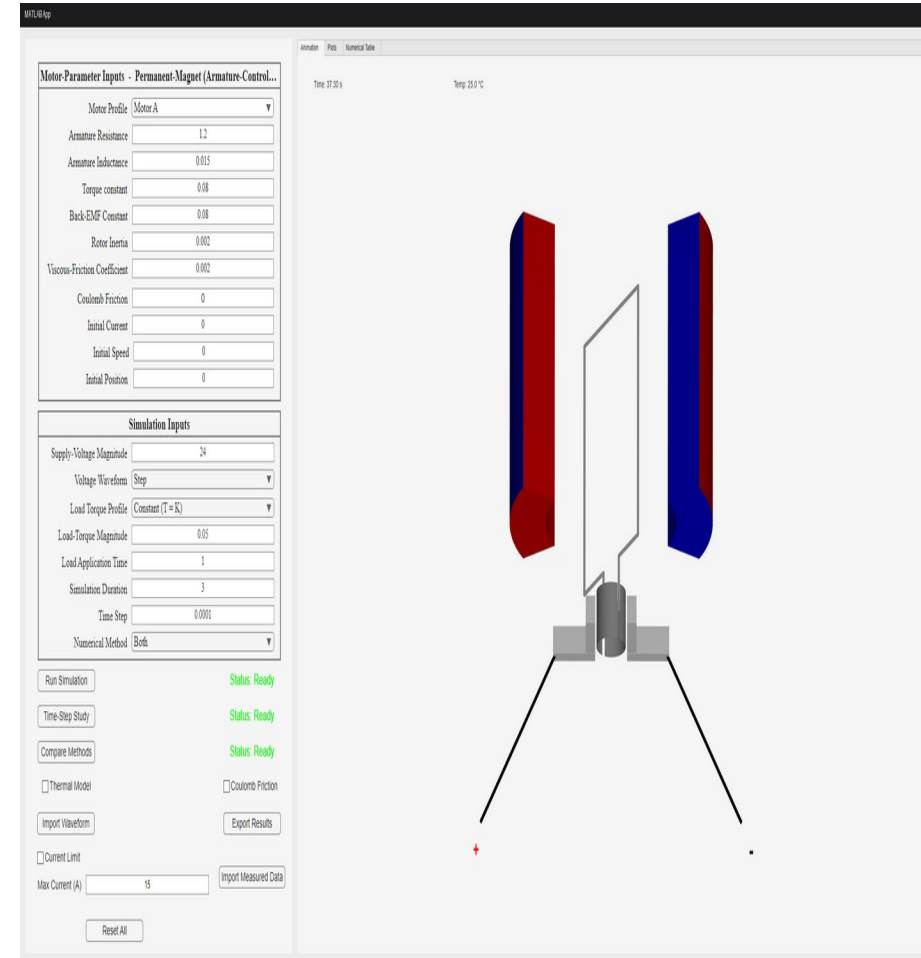
$$k_3 = f(t_n + \frac{h}{2}, \mathbf{x}_n + \frac{h}{2}k_2), \quad k_4 = f(t_n + h, \mathbf{x}_n + hk_3)$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad \mathcal{O}(h^4)$$

Euler: 1 derivative evaluation per step | RK4: 4 evaluations per step, ~30-50x larger h for the same accuracy
Final RK4 stage evaluates its inputs at the left limit of the interval so step inputs do not break 4th-order accuracy

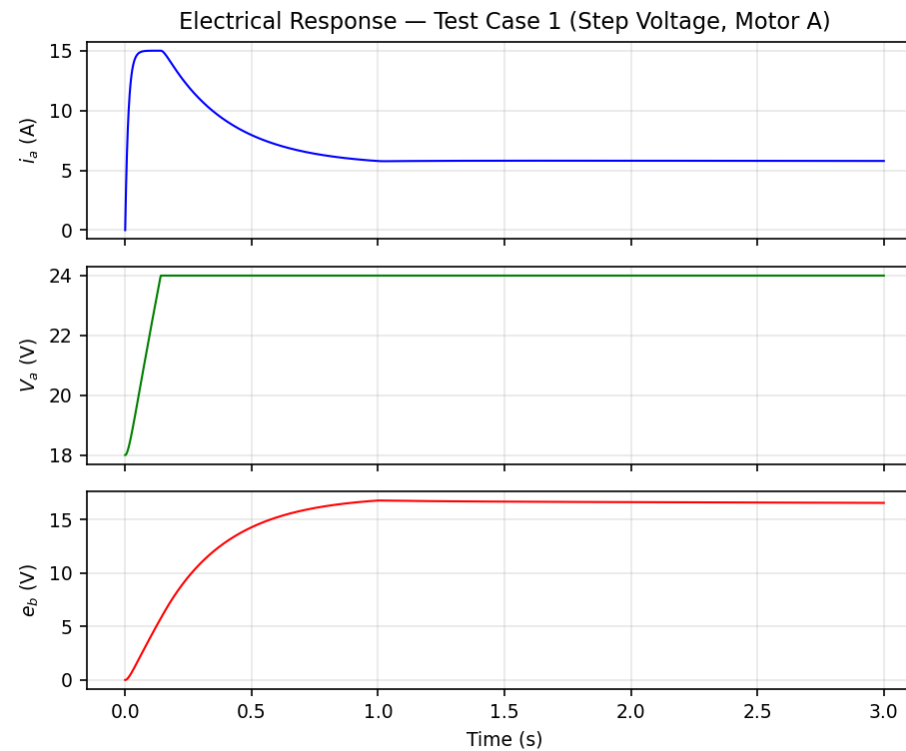
3.3 Design: Simulation Model and GUI

- Left panel: motor parameters and simulation inputs
- Voltage: step, ramp, pulse, imported CSV, user-defined
- Load: constant, linear, quadratic, constant-power, imported
- Tabs: Animation, Plots, Numerical Table
 - Plot View selects Torque-Speed, Electrical, Mechanical or Numerical Analysis
- Controls: Run Simulation, Compare Methods, Time-Step Study, Import, Export, Reset
- Toggles: thermal model, Coulomb friction, current limit



4. Implementation: Test Case 1 - Electrical Response (Current-Limited)

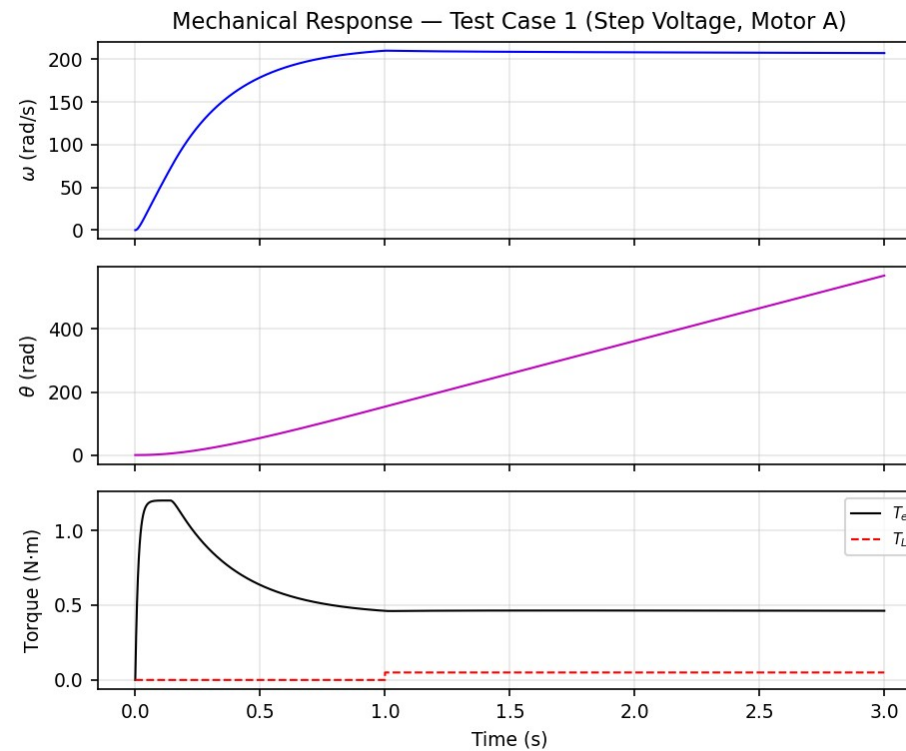
- Motor A, 24 V step at $t = 0$, 0.05 N.m load at $t = 1$ s, 3 s run, $h = 1e-4$ s - current limiter ON (15 A)



15 A limiter clamps the current for ~150 ms; voltage rises 18 V to 24 V as back-EMF builds. (Limiter off: baseline peak is 18.32 A.)

4.1 Implementation: Test Case 1 - Mechanical Response (Current-Limited)

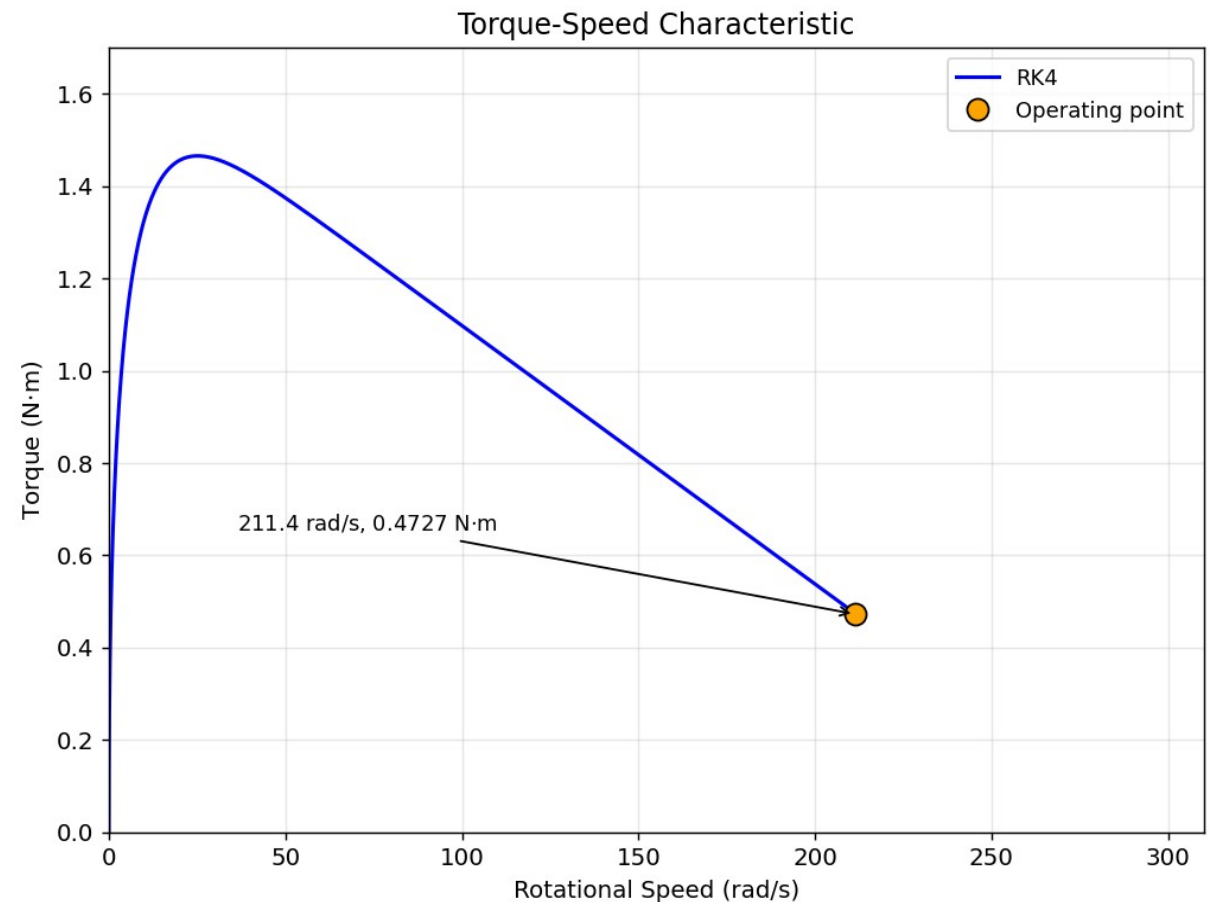
- Speed, shaft position and the torque balance for the same run



Speed settles at 211.4 rad/s - same as with the limiter off, since it only reshapes the transient. Torque absorbs the load step at $t = 1$ s.

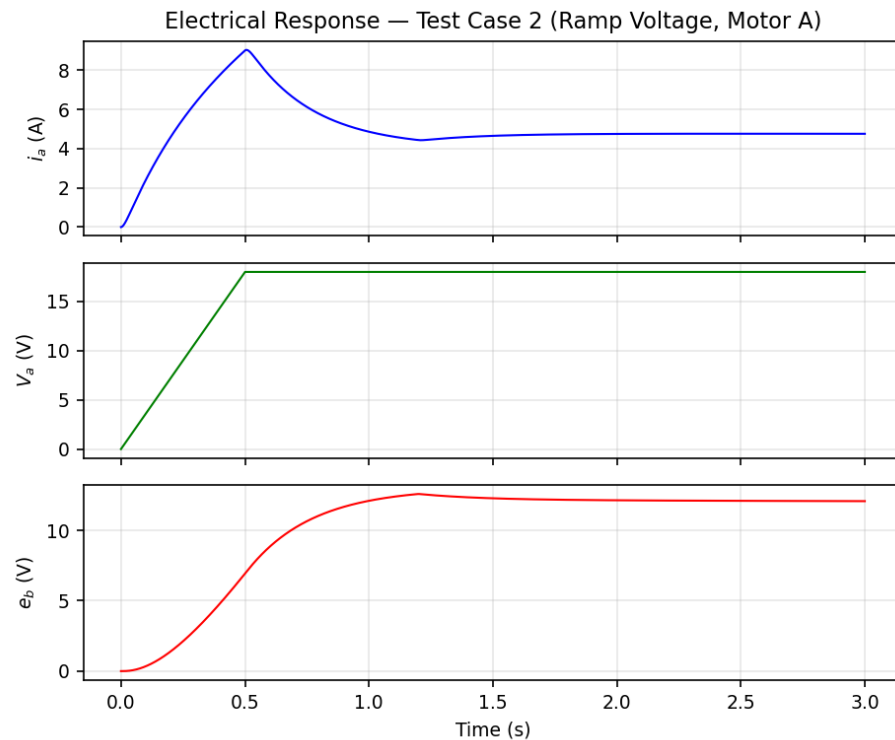
4.2 Implementation: Torque-Speed Characteristic (Baseline, No Limiter)

- Blue curve: the simulated transient trajectory, from standstill to steady state
- Marker: the operating point - where the run settles
- 211.4 rad/s, 0.4727 N·m at that point
- Equals $B\omega + T_L$ at that speed - the torque balance at steady state
 - Verified against the analytical steady state to within 0.01 rad/s



4.3 Implementation: Test Case 2 - Ramp Voltage

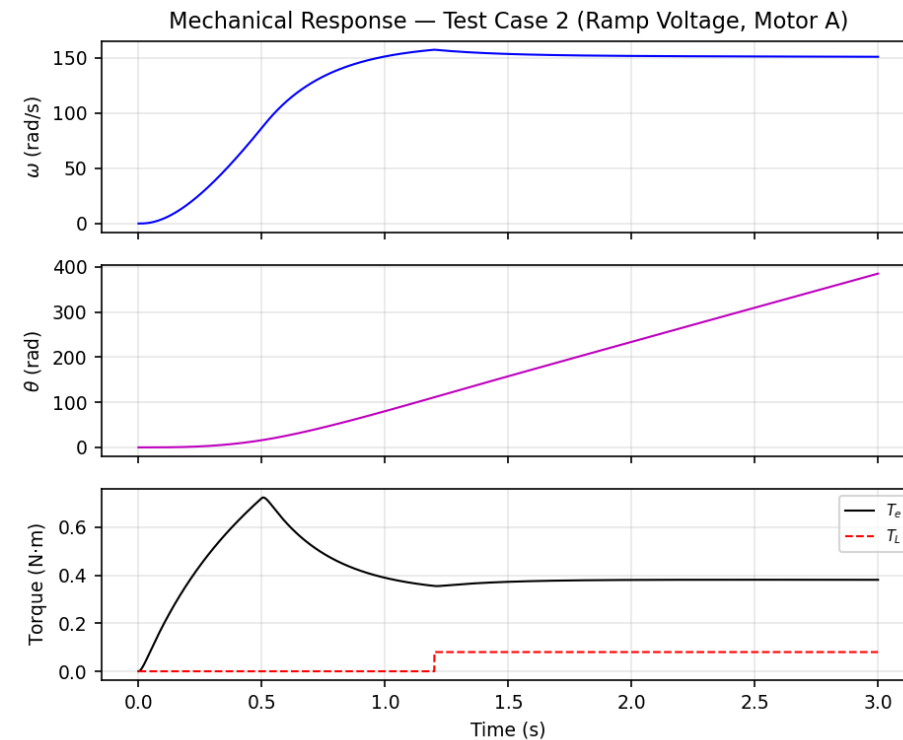
- Motor A, 18 V ramped over 0.5 s, 0.08 N.m load at $t = 1.2$ s, 3 s run, $h = 2e-4$ s



Electrical response

Gradual voltage rise limits the peak current to 9.05 A, so the 15 A limiter never engages

Pulse and imported-waveform cases (Test Cases 3 and 4) are in the full report, Section 4.3.

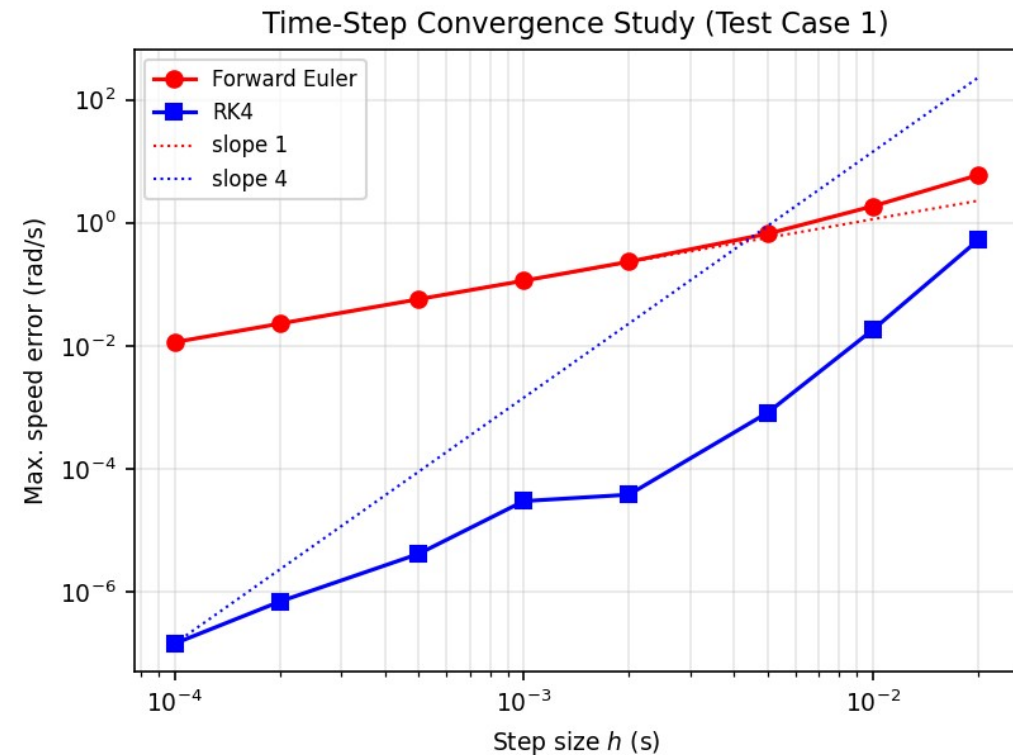


Mechanical response

Speed settles at 151.0 rad/s; the 0.08 N.m load is applied at $t = 1.2$ s

4.4 Numerical Analysis: Method Comparison and Time-Step Study

- Errors measured against a fine-step RK4 reference ($h = 1e-5$ s)
- Euler error follows the expected first-order trend
- RK4 shows the expected fourth-order convergence overall
- At $h = 1e-3$ s: Euler 0.114 rad/s vs RK4 $3.0e-5$ rad/s
- Deviations from ideal 16x scaling occur from reference-error limits and input discontinuities
- Unstable step sizes are detected and excluded, not plotted

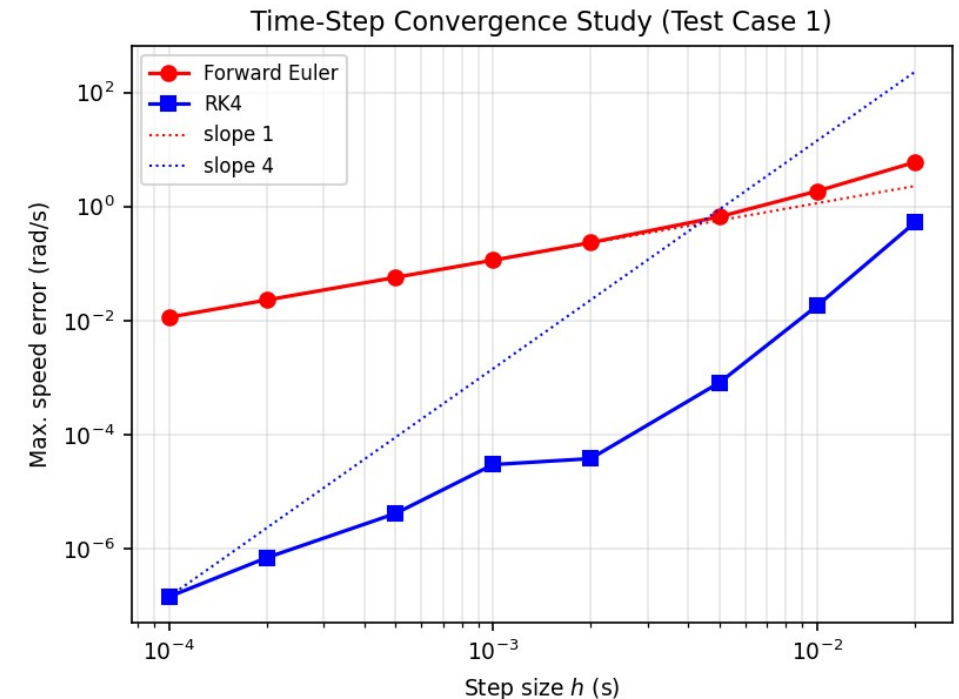


5. Design Analysis and Evaluation

- 5.1 Investigations (PO(d))
- 5.2 Limitations of Tools (PO(e))

5.1 Investigations (PO(d))

- **Design of experiment**
 - Euler and RK4 run at eight step sizes from $1e-4$ to $2e-2$ s
 - Maximum speed error measured against a fine-step RK4 reference
- **Results**
 - Euler: approximately first-order convergence
 - RK4: fourth-order convergence as an overall trend
 - Individual step-to-step error ratios deviate from the ideal 16x scaling
 - Both methods stable across the tested range
- **Finding during development**
 - RK4 silently dropped to first order on steps meeting an input discontinuity
 - Corrected by evaluating the final stage at the left limit of the interval



5.2 Limitations of Tools (PO(e))

- Fixed-step solvers need the user to choose h
 - Motor A electrical time constant is ~ 13 ms
 - Euler diverges above $h \sim 26$ ms, RK4 above $h \sim 36$ ms
 - The Time-Step Study detects divergence and reports it instead of plotting it
- ode45 reference is itself limited by its tolerances (set to $1e-10$ here) and by rounding
- Current limiter is an idealised instantaneous voltage clamp, not a real chopper or PWM stage
- Thermal model is lumped first-order - no spatial temperature gradient in the windings
- Animation speed depends on the host machine redraw rate

6. Future Work

- Regenerative operating mode - returning energy to the supply when overhauled by the load
- Magnetic saturation of the torque and back-EMF constants at high current
- Full closed-loop PWM current-limiting model in place of the idealised voltage clamp
- Closed-loop speed and position control (PI / PID) built on the present plant model
- Export the animation as a video file alongside the existing CSV data export

7. References

- S. C. Chapra and R. P. Canale, Numerical Methods for Engineers, McGraw-Hill.
- S. J. Chapman, Electric Machinery Fundamentals, McGraw-Hill.
- MathWorks, MATLAB Ordinary Differential Equation Documentation.
- MathWorks, MATLAB App Designer Documentation.
- EEE 212 Numerical Techniques Laboratory, MATLAB Project Handbook, Dept. of EEE, BUET.